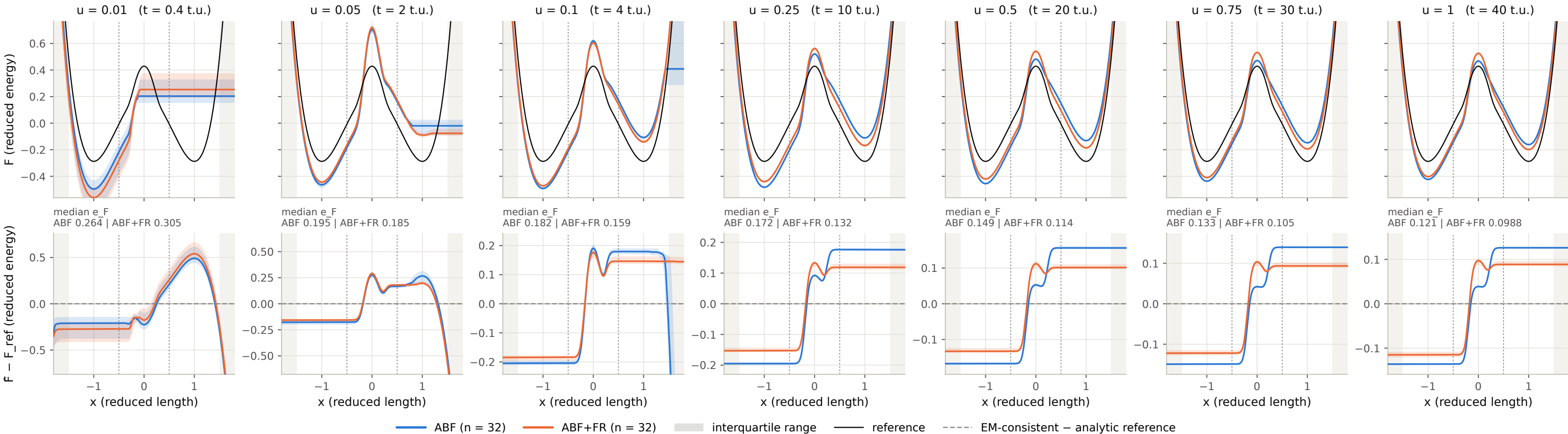


# Gateway family matched\_free\_energy/shift · N = 2048 · T\_N = 40 t.u.

n\_steps = 1,600,000 · h = 2.5e-05 · B = N n\_steps = 3.277e+09 walker-steps · complete seeds: ABF 32/32, ABF+FR 32/32

Free-energy profile snapshots  $F_t$  vs reference (top) and difference (bottom)



Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC\_PLAN section-4 quantities, unsmoothed.

Profile = the accepted scorer's read-out (exact integral of  $\Gamma = M / (C + 1)$  at the 181 grid nodes), centred on the eval window like  $F_{ref}$ : the alignment  $e_F$  uses, identical for every method, seed and N. Top: profiles; bottom: difference (y-scale per column).

Dashed guides: region boundaries  $x = \pm 0.5$  (left well / gate / right well); shaded  $|x| > 1.5$ : outside the scored eval window. Top-row y range = the reference over the scored window  $\pm 35\%$  of its span (larger excursions are clipped there and show in the difference row).

No smoothing.

eqb\_plot\_config/2 · config shift.json · 2026-10-10 14:20 UTC